DATA STRUCTURES

Easy-Level Tech Quiz · Undergraduate (UG)

Name: _____ **Class:** _____

Instructions: Choose the best answer for each question.

1. Which data structure follows LIFO?

A. Queue
B. Stack
C. Tree
D. Graph

2. Which data structure follows FIFO?

A. Stack
B. Tree
C. Queue
D. Graph

3. Which data structure uses nodes and pointers?

A. Linked List
B. Array
C. Integer
D. String

4. What is the first index of an array in C++?

A. 1
B. -1
C. 0
D. 2

5. Which operation adds an element to a stack?

A. Pop
B. Push
C. Delete
D. Search

6. Which operation removes the top element from a stack?

A. Push
B. Insert
C. Pop
D. Enqueue

7. Which operation adds an element to a queue?

A. Dequeue
B. Push
C. Enqueue
D. Pop

8. Which data structure is used in BFS?

A. Queue
B. Stack
C. Array only
D. Tree only

9. Which data structure is commonly used in DFS?

A. Queue only
B. Stack
C. Hash table only
D. Array only

10. A node in a linked list contains data and what else?

A. Pointer or link
B. Compiler
C. Loop
D. Keyword

11. Which data structure represents hierarchical data?

A. Queue
B. Stack
C. Tree
D. Array

12. Which data structure consists of vertices and edges?

A. Stack
B. Graph
C. Queue
D. Array

13. Which of these is a linear data structure?

A. Graph
B. Tree
C. Array
D. None of these

14. Which sorting algorithm repeatedly compares adjacent elements?

A. Merge Sort
B. Bubble Sort
C. Binary Search
D. DFS

15. Which algorithm finds an element in a sorted array by halving the search range?

A. Linear Search
B. Bubble Sort
C. Binary Search
D. BFS

16. What is the time complexity of accessing an array element by index?

A. O(n)
B. O(log n)
C. O(1)
D. O(n^2)

17. Which data structure is suitable for undo operations?

A. Queue
B. Stack
C. Graph
D. Tree

18. What is the maximum number of children a binary tree node can have?

A. 1
B. 2
C. 3
D. 4

19. Which data structure is commonly used to manage function calls?

A. Stack
B. Queue
C. Graph
D. Heap only

20. What does an empty queue contain?

A. One element
B. Two elements
C. No elements
D. One node always

ANSWER KEY

Use this section for the quiz organizer or for checking answers.

1. B. Stack
2. C. Queue
3. A. Linked List
4. C. 0
5. B. Push
6. C. Pop
7. C. Enqueue
8. A. Queue
9. B. Stack
10. A. Pointer or link
11. C. Tree
12. B. Graph
13. C. Array
14. B. Bubble Sort
15. C. Binary Search
16. C. O(1)
17. B. Stack
18. B. 2
19. A. Stack
20. C. No elements